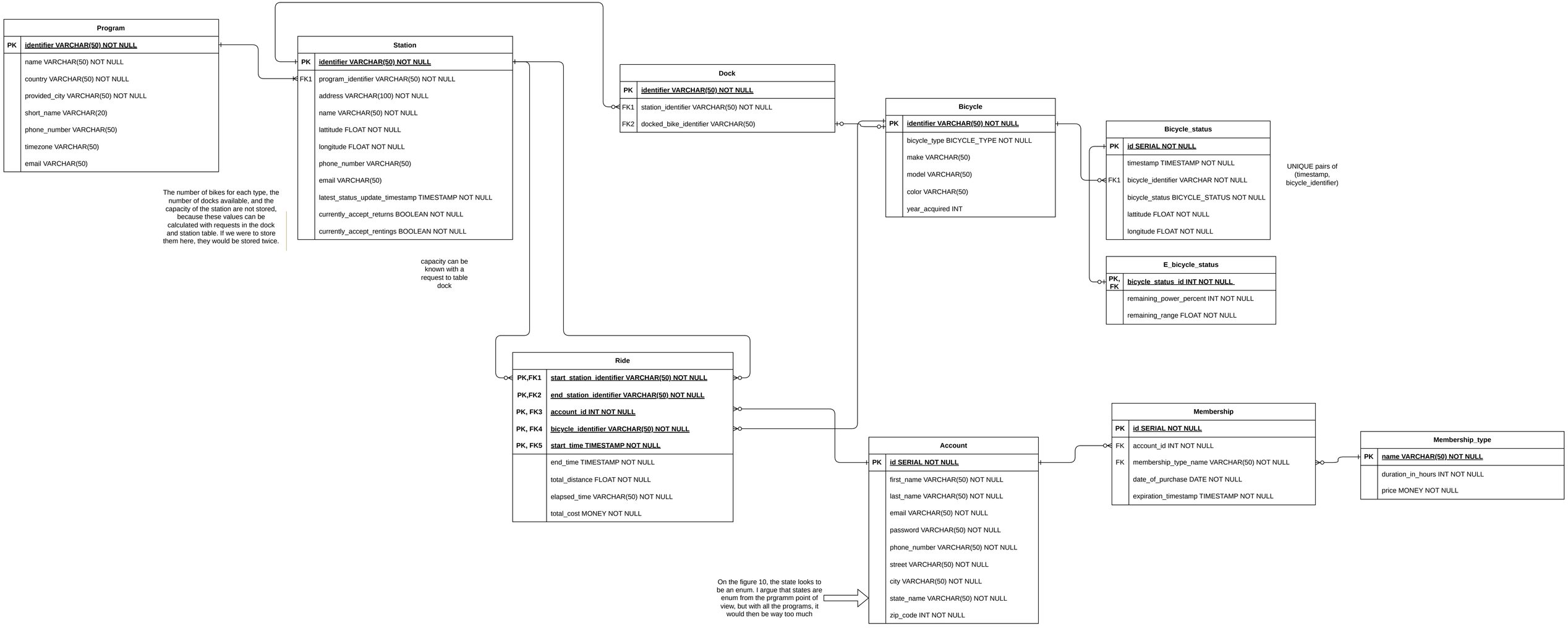
Program
PK identifier VARCHAR(50) NOT NULL
name VARCHAR(50) NOT NULL
country VARCHAR(50) NOT NULL
provided_city VARCHAR(50) NOT NULL
short_name VARCHAR(20)
phone_number VARCHAR(50)
timezone VARCHAR(50)
email VARCHAR(50)
Station
PK identifier VARCHAR(50) NOT NULL
FK1 program_identifier VARCHAR(50) NOT NULL
address VARCHAR(100) NOT NULL
name VARCHAR(50) NOT NULL
lattitude FLOAT NOT NULL
longitude FLOAT NOT NULL
phone_number VARCHAR(50)
email VARCHAR(50)
latest_status_update_timestamp TIMESTAMP NOT NULL
currently_accept_returns BOOLEAN NOT NULL
currently_accept_rentings BOOLEAN NOT NULL
The number of bikes for each type, the number of docks available, and the capacity of the station are not stored, because these values can be calculated with requests in the dock and station table. If we were to store them here, they would be stored twice.
capacity can be known with a request to table dock
Dock
PK identifier VARCHAR(50) NOT NULL
FK1 station_identifier VARCHAR(50) NOT NULL
FK2 docked_bike_identifier VARCHAR(50)
Bicycle
PK identifier VARCHAR(50) NOT NULL
bicycle_type BICYCLE_TYPE NOT NULL
make VARCHAR(50)
model VARCHAR(50)
color VARCHAR(50)
year_acquired INT
Bicycle_status
PK id SERIAL NOT NULL
timestamp TIMESTAMP NOT NULL
FK1 bicycle_identifier VARCHAR NOT NULL
bicycle_status BICYCLE_STATUS NOT NULL
lattitude FLOAT NOT NULL
longitude FLOAT NOT NULL
UNIQUE pairs of (timestamp, bicycle_identifier)
E_bicycle_status
PK, FK bicycle_status_id INT NOT NULL
remaining_power_percent INT NOT NULL
remaining_range FLOAT NOT NULL
Ride
PK,FK1 start_station_identifier VARCHAR(50) NOT NULL
PK,FK2 end_station_identifier VARCHAR(50) NOT NULL
PK, FK3 account_id INT NOT NULL
PK, FK4 bicycle_identifier VARCHAR(50) NOT NULL
PK, FK5 start_time TIMESTAMP NOT NULL
end_time TIMESTAMP NOT NULL
total_distance FLOAT NOT NULL
elapsed_time VARCHAR(50) NOT NULL
total_cost MONEY NOT NULL
Account
PK id SERIAL NOT NULL
first_name VARCHAR(50) NOT NULL
last_name VARCHAR(50) NOT NULL
email VARCHAR(50) NOT NULL
password VARCHAR(50) NOT NULL
phone_number VARCHAR(50) NOT NULL
street VARCHAR(50) NOT NULL
city VARCHAR(50) NOT NULL
state_name VARCHAR(50) NOT NULL
zip_code INT NOT NULL
On the figure 10, the state looks to be an enum. I argue that states are enum from the prgramm point of view, but with all the programs, it would then be way too much
Membership
PK id SERIAL NOT NULL
FK account_id INT NOT NULL
FK membership_type_name VARCHAR(50) NOT NULL
date_of_purchase DATE NOT NULL
expiration_timestamp TIMESTAMP NOT NULL
Membership_type
PK name VARCHAR(50) NOT NULL
duration_in_hours INT NOT NULL
price MONEY NOT NULL